

This exam is for:

ECSE-6050

Flip this page over only when instructed to do so.

This exam is open book, open notes. You may use your computer/tablet only to view course content. This includes and is limited to: the textbook, lecture notes (both personal and provided), homeworks, homework solutions, and provided back exam(s). All other uses are prohibited. Typing on the computer/tablet is not allowed; excluding for scrolling through documents.

For each problem: mark the problem number and all subquestions clearly and box your final answer(s). Answers that are not supported by work will not receive credit. Questions with symbolic answers are not expected to be significantly simplified unless specified.

All answers will be written on blank paper, single-sided.

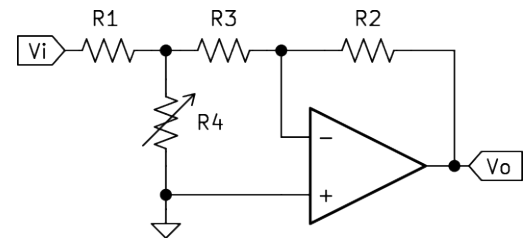
Write on one side of answer paper only.

Answers that are not supported by work will not receive credit
Write on one side of your answer paper only

Q.1 (70 pnts): Answer the subquestions below with respect to the circuit on the right. Assume that $r_d = R$, $r_o \neq 0$, and $\alpha \neq \infty$ unless otherwise stated or implied. *Note: This circuit is effectively one-half the circuit used in homework #1, book problem 1.27, where R_4 in this version is used to control the gain.*

- Find $A_{ideal} = v_o/v_i$. Fully simplify (10).
- Find the input impedance, R_i under ideal conditions. (5)

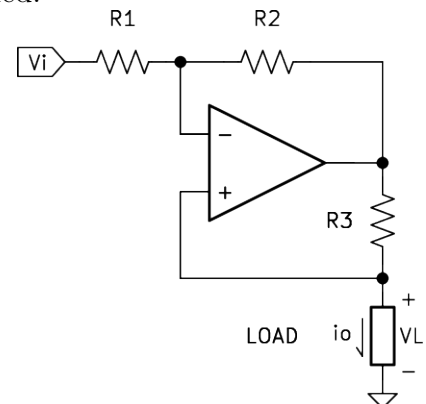
For all remaining subquestions, assume R_1 - R_4 are equal to R . Note that $r_d = R$ is already specified.



- Find A using exact analysis with $r_o = 0$. (10)
- Calculate the return ratio, T . (15)
- Calculate the feedthrough gain, a_{ft} . (10)
- Find the non-ideal A via return ratio analysis. (5)
- Calculate the circuit impedance experienced by R_3 : R_{c3} . Simplify the resulting equation assuming $r_o \ll R$ and $a \rightarrow \infty$. (10)
- Using answers from one-or-more subquestions above, find the effective error amplifier gain, a_ϵ . If you do not have answers for the required subquestions, describe clearly how you would find a_ϵ if you had, indicating the answers that would have been used. (5)

Q.2 (20 pnts): Answer the subquestions below with respect to the voltage-to-current amplifier circuit on the right. Assume ideal conditions unless otherwise stated or implied.

- Find i_o such that it is in the standard form $i_o = Av_i - v_L/R_o$. Clearly identify the values for A and R_o . Assume ideal conditions. (10)
- The op-amp is powered via ± 10 V supplies and has a quiescent current of 2 mA. With $R_1, R_2, R_3 = 1$ k Ω , what is the power consumed by the amplifier if $V_L = 2$ V and $i_o = 4$ mA? (10)



Q.3 (10 pnts): A differential amplifier has a gain of 100 V/V and a CMRR of 80 dB. With $v_1 = 2.47$ V and $v_2 = 2.56$ V, what is actual output and output error as compared to ideal due to common-mode amplification? Answer in volts.